

# Recap Checkpoint 1 — after 1.1 (Processors, Input/Output & Storage) + 1.2 (Software)

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Name: \_\_\_\_\_ Date: \_ **Mark:** \_\_\_\_\_ / 34

*OCR H446 cumulative recap — mixes every topic taught so far. Show all working.*

## Questions

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**Q1.** State the function of the **Program Counter (PC)** and the **Current Instruction Register (CIR)** during the Fetch–Decode–Execute cycle. **[2]** (AO1) — 1.1.1

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**Q2.** A processor uses **pipelining**. (a) Explain what is meant by pipelining. **[2]** (AO1)

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(b) State **one** situation in which the pipeline may have to be flushed. **[1]** (AO1)

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**[3 total]** — 1.1.2

**Q3.** A games studio is choosing hardware to render graphics that apply the **same calculation across millions of pixels in parallel**. Explain why a **GPU** is better suited to this than a general-purpose CPU. **[3]** (AO2) — 1.1.2

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**Q4.** A digital camera stores photographs onto a removable memory card. (a) Name the **type of secondary storage** used by a typical memory card. **[1]** (AO1)

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(b) Justify why this storage type is more suitable for a camera than a magnetic hard disk drive, referring to **two** distinct properties. **[2]** (AO2)

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**[3 total]** — 1.1.3

**Q5.** Explain how an **interrupt** is handled by the operating system, referring to the role of the **interrupt service routine (ISR)** and the **stack**. **[4]** (AO1) — 1.2.1

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**Q6.** An operating system uses a **paging** approach to memory management together with **virtual memory**. (a) Explain what is meant by virtual memory. **[2]** (AO1)

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(b) Describe **one** drawback of relying heavily on virtual memory. **[1]** (AO2)

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**[3 total]** — 1.2.1

**Q7.** A developer writes a program in a high-level language. (a) State **one** difference between a **compiler** and an **interpreter**. **[2]** (AO1)

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(b) Place the four stages of compilation in the correct order: *code generation*, *lexical analysis*, *syntax analysis*, *optimisation*. **[2]** (AO1)

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**[4 total] — 1.2.2**

**Q8.** A software team is choosing between the **Waterfall** and **Agile (e.g. extreme programming)** methodologies for a project whose requirements are expected to change frequently. Recommend a methodology and justify your choice, referring to **two** characteristics of the chosen methodology. **[4] (AO2)**  
— 1.2.3

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**Q9.** Object-oriented programming uses **encapsulation** and **inheritance**. (a) Explain what is meant by **encapsulation**. **[2] (AO1)**

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(b) Give **one** benefit of **inheritance** to a programming team. **[1] (AO2)**

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**[3 total] — 1.2.4**

**Q10.** A retailer is replacing its ageing tills. The new tills must run reliably for long shifts, recover quickly from power loss, and let staff carry out card payments and stock look-ups simultaneously. The IT manager must decide on the **type of processor architecture (CISC vs RISC)** and the **type of operating system** (real-time, multi-tasking, distributed). Evaluate suitable choices for **both** decisions, justifying each against the retailer's requirements. **[5] (AO3) — synoptic 1.1 + 1.2**

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